Appendices
Pollinator decline reveals uneven impacts of climate-driven biodiversity loss

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Appendix 1. Extended Methods, Materials and Data Processing

All data and codes are available at: https://github.com/gabrielemansi/Macroeconomic-impacts-of-pollination-loss-in-the-EU.git

Baseline pollination scenarios

To build the pollination potential baseline (2020-2070), the initial level of Pollination Potential for the year 2020 relies on data from the INCA (2018) potential (Vallecillo et al., 2022), normalized to a 0-1 scale. The KIP-INCA framework (Vallecillo et al., 2018; Maes et al., 2020) developed a set of ecosystem ser-vices accounts (Maes et al., 2020) including crop pollination. The service is provided by the functional
match between pollination potential, defined as the ecosystem's ecological suitability to support wild insect pollinators, and pollination demand, defined as the spatial extent of pollinator-dependent crops (the use area).
For this project we used estimates of pollination potential provided in "Ecosystem services accounts: maps for 7 ecosystem services produced in KIP INCA 2018 updates" (Vallecillo et al., 2022), available at (https://data.jrc.ec.europa.eu/dataset/4cbd7c1e-6512-4ebe-8ca5-e08209cc3efb / "\CROP_POLLINATION\potential\potential_categories.tif") .

The data is provided in a GeoTIFF images with the following characteristics:
Size: Image size: 4009, 4045 - Pixel Size: 1000, -1000
CRS: EPSG:3035 - ETRS89-extended / LAEA Europe - Projected
Extent: Corner Coordinates:

- Upper Left 2549000.0000000000000000,5447000.0000000000000000
- Lower Left 2549000.0000000000000000,1402000.0000000000000000
- Upper Right 6558000.0000000000000000,5447000.0000000000000000
- Lower Right 6558000.0000000000000000,1402000.0000000000000000

Unit : meters
The file includes the following bands: Year 2000, Year 2006, Year 2012, Year 2018

- min value: 1
- max value: 4

Description: Pollination potential is expressed with a categorical variable (1=None, 2=low, 3=medium, 4=high). The assessment of pollination potential is based on an indicator of the environmental suitability to support wild insect pollinators. It integrates two different models: an Expert-Based Model for solitary bees and a Species Distribution Model for bumblebees.

Vallecillo et al., 2018 data classifies EU land surface into categories of pollination potential (1=None, 2=low, 3=medium, 4=high) based on a highly detailed assessment of land-cover composition and land-use patterns, aimed at identifying habitat elements suitable for pollinators. The assessment is based on an indicator of the environmental suitability to support wild insect pollinators which integrates two different models: an Expert-based Model (EBM) and a Species Distribution Model (SDM). More specifically, they build upon previous work undertaken by European Commission Joint Research Centre staff, which has resulted in an EBM with a spatial resolution of 1 ha (100 x 100 m grid-cell) (Zulian et al., 2013), and a SDM based on bumblebee records, with a spatial resolution of 100 km2 (10 x 10 km) (Polce et al., 2013). Each of these approaches has some strengths and weaknesses: the EBM for instance, has the advantage of being able to account for the effect of detailed local information, such as the presence of wild flower edges between crop-fields, or other small patches of habitat suitable for

pollinators. The EBM, however, might fail to reflect the environmental suitability for poorly known species, or to capture environmental characteristics that can modify the expected suitability (e.g. climatic differences) or, again, it might not be able to predict species richness. The SDM, on the other hand, has the advantage of being informed by actual species records, but it is constrained by the spatial and temporal resolution of these records. Hence, the SDM might fail to capture the effect of local landscape elements, if their accuracy is greater than what is available for the species records. By integrating the EBM and the SDM approach, therefore, they are able to better reflect the environmental suitability to support wild insect pollinators (and, hence, the pollination potential). Both models provide a 'suitability score' between 0 and 1 for each grid-cell. The 'suitability' is often interpreted as the 'capacity of the environment to support insect pollinators' (EBM) or the 'probability of occurrence of insect pollinator' (SDM). The EBM-suitability is based on experts' knowledge of the species ecology, solitary bees in our case. The SDM-suitability on the other hand, is derived through, e.g., statistics or machine learning techniques, which are used to characterise the 'quality' of the environment where species are recorded. In simple words, within a SDM, the relations between the environmental variables characterising the species' sightings, bumblebees in our case, are used to predict the environmental suitability across the area of interest. Since the SDM is informed by the sightings of species, it can capture complex relations among the different variables characterising the environment (e.g. land use and land cover, climate, etc.) where species are found. Environmental suitability is interpreted as a proxy for the 'probability of occurrence' for a species, therefore, by defining a threshold it is able to distinguish potential presence from absence (Liu et al., 2005). The European Commission Joint Research Centre has produced individual SDMs for 47 bumblebee species, as well as a SDM resulting from aggregating all single SDMs. The score of the aggregated SDM is the mean of the single species' model for all species predicted to be present in a given cell, with ~ 76 ~ equal weights across species (weight =1/N, where N is the number of species present in a given cell). The original models by Zulian et al. (2013) and Polce et al. (2013) were adapted to meet the requirements of the accounting, such as the need to rely on datasets regularly updated. The outputs of the updated models were then averaged to estimate the potential availability of wild insect pollinators to relevant crop groups: the pollination potential.

Theme	Year		
	2000	2006	2012
LULC – Dynamic dataset	CORINE Accounting Layer 2000	CORINE Accounting Layer 2006	CORINE Accounting Layer 2012
Roads – Static dataset	Road network from TeleAtlas 2006 version (major roads only, corresponding to road category 0, 1 and 2); if a more recent version becomes available, it can be used from 2012 onward.		
Climate data – Dynamic dataset	Gridded Meteorological data from Agri4Cast[17] (Mean air temperature and total global radiation used for the Expert-Based model) and E-OBS[18] (minimum and maximum air temperature, sum of precipitation are used to compute the bio-climatic variables used for the Species-Distribution model - These variables were already computed and during the model calibration phase, and the most relevant one were selected.		
Species records – Static dataset	Bumblebee records from Atlas Hymenoptera[19]: • 47 species selected from ca. 60 (excluded species with very few records) • 10 x 10 km grid • 1991 to 2014		

Figure A1: Input data for the spatial indicator of pollination potential
Source: Vallecillo et al., 2018

The final model output is an indicator of environmental suitability that is used to delineate service providing areas (SPA) with different suitability for pollinators: (1) High: environmental suitability above 0.3 ; (2) Medium: environmental suitability between 0.3 and 0.2; (3) Low: environmental suitability between 0.2 and 0.1; (4) None: environmental suitability below 0.1 .

To ensure consistency of our assessment with respect to other literature, we compared the Pollination Potential used in our assessment (INCA 2018 normalized to 1) against the 2015 Pollination Sufficiency baseline from von Jeetze et. al 2023 (variable "pollSuff_2015_Ref_10s.tif").
Their study employs the coupled MAgPIE-SEALS modeling system to assess fine-scale changes in pollination sufficiency and soil loss from water erosion under four exploratory (what if) land-use scenarios projected for 2050. Pollination Sufficiency is defined as the amount of semi-natural habitat within typical foraging distances of wild pollinator communities surrounding cropland, and represents a proxy for both wild pollination supply and configurational landscape heterogeneity, an important driver of biodiversity and other regulating ecosystem services such as biological pest control. Thy used a modeling framework which integrates spatial and ecological parameters to simulate interactions between pollinator habitats and crop dependencies over time. Calculations are performed at a 300 m × 300 m resolution using a landscape-based indicator based on two components:

- Habitat Definition: Includes forest, non-forest vegetation, and grassland—assumed to support wild pollinators.
- Foraging Radius: A 2 km buffer is applied around each cropland pixel. If ≥30% of this area is semi-natural habitat, the pixel receives a sufficiency score of 1. If below 30%, the score is scaled linearly between 0 and 1.

These scores quantify pollination supply and landscape heterogeneity, with each cropland pixel assigned a value from 0 to 1:

- 1 = sufficient habitat (≥30%)
- (0 – 1) = proportional habitat (<30%)

Scenario Design and Assumptions The reference scenario—Business-as-Usual (BAU)—follows the SSP2 "middle-of-the-road" pathway, assuming moderate population and income growth alongside current national land conservation policies. Climate data are used for baseline biophysical simulations but do not factor into future pollinator dynamics; climate change impacts such as altered rainfall or temperature are excluded from the pollination modeling.
In the baseline data (base year 2018 and 2015), the two metrics proved to be broadly comparable, with closely aligned mean values across the EU.

Layer	Mean	Dataset
1 2018	0.64871	INCA_2018
2 2015	0.73535	von Jeetze

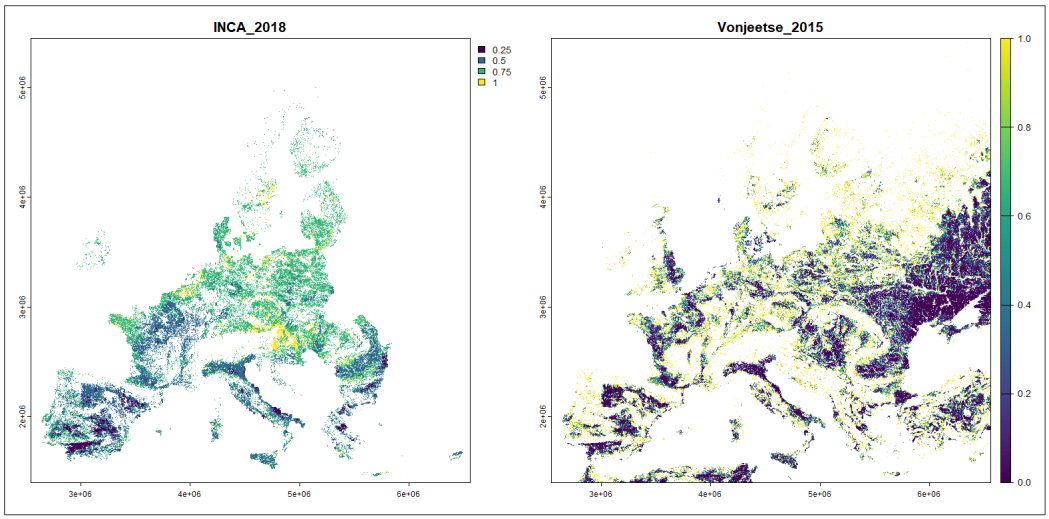


Figure A2: Comparison between von Jeetze et. al 2023 pollSuff_2015_Ref_10s and INCA_2018 pollination potential normalized to 0-1.

The INCA pollination potential represents a snapshot of current ecological conditions and does not explicitly incorporate temporal trends. Likewise, species distribution models (SDMs) generally omit several relevant stressors, including pollutant exposure (see Appendix 1). Introducing an explicit temporal degradation complements the static baseline and better captures observed declines not otherwise represented in the modelling framework.

Future changes are modelled by scaling down the 2020 baseline potential up to 2050 in line with observed historical trends in pollination loss. A -1% annual decline in pollination potential is imposed up to 2035, under the assumption that the observed negative trajectory persists throughout the present decade and reaches its maximum intensity around that year. From 2035 onwards, the rate of decline is assumed to decelerate progressively, following a linear tapering path until it converges to a 0% change rate by 2050. Prior to imposing future climate change effects, no further decline is projected between 2050 and 2070, implying a stabilisation of pollination potential in the long run under the baseline scenario. The assumption is deliberately conservative, intended to avoid overstating long-term losses under substantial uncertainty in ecological trajectories and policy effectiveness.

In particular, it reflects the fact that existing policy interventions, such as the EU Pollinators Initiative and earlier reforms under the Common Agricultural Policy (CAP), may not yet have fully translated into observable trends. Extrapolating current declines over extended horizons would otherwise rely on a strong and potentially unjustified assumption of persistence, with a risk of systematic overestimation of long-run impacts.

The pollination potential assessed in this way is used as the baseline for the no-climate-change scenario (which holds future climate constant), while recognising that the observed trend on which it is calibrated is not fully climate-independent (Colom et al., 2025); the resulting overlap with the climate channel carried by the SDM ensemble is treated as a scenario dimension rather than assumed away. The degradation is imposed uniformly across space, so that all spatial heterogeneity in the projected pollination potential originates from the SDM ensemble; the reasons for this are set out in the dedicated subsection below.

The broader body of literature reviewed (see: Appendix 2), together with expert consultation with the INCA team at the Joint Research Centre (JRC), supports the adoption of a -1% annual decline as a robust and defensible assumption for the EU. This parameter is consistent with estimates developed within the EU Science and Technology for Pollinating Insects (STING) project (Potts et al., 2024),

ensuring methodological coherence with the most recent EU-level scientific assessments and reinforcing its empirical plausibility within the modelling framework.

The -1% yr baseline rate is anchored on three independent pan-European lines of evidence.

First, the EU Grassland Butterfly Index, computed from standardised transect monitoring across all 27 Member States, declined by 47% over 1991-2024 (-1.9% yr) in the latest update (Eurostat, 2025; Van Swaay et al., 2025 b), while the 1991-2020 window yields -1.23% yr. This is a widely used proxy for pollinator-related biodiversity change due to its long temporal and spatial coverage and sensitivity to the same environmental pressures that affect wild pollinators. These include land-use change, habitat loss and fragmentation (resulting, among other factors, from agricultural intensification and monocultures, urbanization and infrastructure development, removal of hedgerows and field margins, loss of semi-natural grasslands, and decline of wild flowering plants), as well as exposure to environmental pollutants such as pesticides (especially systemic insecticides) (Nicholson et al., 2024) and fungicides, and the presence of invasive species, all factors that are more prevalent in intensive agricultural systems. Since INCA pollination potential and SDMs in general do not account for many of these sources of impact (such as pollutants), imposing historical trend over baseline future pollination allows us to capture real-world pollination decrease trends not specifically modelled both in the initial data and in the SDM projections used for climate impacts. In this context, the 1991–2020 trend is used to validate the no-climate-change baseline trend, on the understanding that it is only partially independent of climate-driven effects (Colom et al., 2025; Van Swaay et al., 2025b). The imposed -1% yr is accordingly set below the -1.23% yr observed over the same window, the difference representing a low-boundary estimate of the observed value, partial netting-out of the possible climate-component when projecting the monitoring signal.

Second, the European Red List assessments provide independent, taxon-resolved corroboration of continent-wide decline: 9% of bee species were assessed as threatened with negative population trends for over 45% of species and 57% Data Deficient (Nieto et al. 2014), a picture substantially worsened in the 2026 update, in which nearly 100 additional wild bee species were classified as threatened and over 20% of bumblebees now face extinction risk (Michez et al. 2026); 37% of the 890 European hoverfly species are threatened, with 244 species affected by climate-driven habitat degradation and increased fire frequency (Vujić et al. 2022); and threatened European butterfly species increased by 76% over the last decade, with 28.3% of species threatened or Near Threatened (Van Swaay et al., 2025a). These assessments rest on a harmonised, taxonomically validated occurrence base covering 97% of the European bee fauna (2,083 of 2,138 species) and 97% of the hoverfly fauna (886 of 913), assembled and expert-validated under the Safeguard, ORBIT, Taxo-Fly and PULSE projects (Sentil et al., 2026; described in Appendix 2), which constrains the extent to which continent-wide decline signals can be attributed to taxonomic or geographic artefacts.

Third, the ensemble reconstruction of Ghisbain et al. (2024) demonstrates coherent declines in the ecological suitability of Europe for 46 bumblebee species since 1900, based on 401,046 georeferenced occurrence records, providing a spatially explicit historical benchmark on the same modelling framework used for our GHSB climate scenarios. This line corroborates the direction and order of magnitude of long-run decline but does not constitute evidence independent of the impact channel; it is reported here as an internal consistency check rather than as a third independent anchor.

The imposed rate is also consistent with the expected long-term trend of agro-ecosystem loss rate reported by the EU (2023, a, b), which estimates ecosystem loss in agro-ecosystems at 1.5% per year in the absence of further action, and proxies the expected decline of those services relying on the ecosystem's status and trend (see Appendix 2).

Two qualifications follow. First, the observed monitoring trend is not completely climate-independent, Colom et al. (2025) attribute a component of European butterfly population declines directly to climate warming, using the same eBMS network from which the Grassland Butterfly Index is derived, imposing

the full observed rate on the no-climate-change baseline would double-count the climate channel already carried by the SDM ensemble.
We therefore treat the baseline rate as a modelling assumption and scenario dimension rather than a fixed parameter, with the -1% yr case representing an lower bound of observed trend. Second, the assumption of progressive deceleration to 2050 is consistent with evidence that pollinator richness declines in Europe were steepest before 1990 and became substantially less accentuated, or partially reversed, in the following decades in regions where land-use intensification had largely run its course (Carvalheiro et al. 2013). That evidence is however restricted to north-western Europe, no equivalent analysis being available for southern or eastern Europe given the data shortfalls documented below, so the deceleration assumption is extrapolated rather than observed across the full EU27. Conversely, the monitoring record probably captures only part of the historical decline, since many grassland populations were extirpated before systematic monitoring began: in the Netherlands, butterfly distributions have contracted by more than 80% since 1890 (Van Strien et al., 2019; Van Swaay et al., 2025b). Taken together, these considerations support treating the imposed rate as a conservative central case rather than a best estimate.

Spatial uniformity of the baseline degradation.
The degradation described above is imposed uniformly across space. This reflects a data-availability constraint rather than an assumption of spatially homogeneous pressure: no harmonised, effort-corrected time series of wild pollinator status is currently available below the EU aggregate. Two candidate sources exist, and will maybe provide better detail in future.
The first is standardised transect monitoring. The eBMS database integrates 38 national and subnational Butterfly Monitoring Schemes across 33 countries and records from more than 16,000 standardised transects, of which over 6,000 within the EU27 contribute to the Grassland Butterfly Index for 17 grassland species (Van Swaay et al., 2025b). Species annual collated indices are computed per constituent scheme as an intermediate step of the indicator workflow, but only the EU27 aggregate is publicly released, as a single series of 34 annual values. Three features prevent the published indicator from being disaggregated as it stands. Constituent schemes differ markedly in their starting year, several Mediterranean and central-eastern schemes having been established only around 2019–2020; the EU27 series is completed by carrying forward the last non-missing index and by anchoring short series to the indicator value in their first year, so that national coverage is in part imputed rather than observed; and several schemes are subnational (Catalonia, the Basque Country, Flanders and Wallonia are separate schemes) so that the mapping from scheme to Member State is not one-to-one. Harmonised national Grassland Butterfly Indices are supposed to become available through the National Restoration Plans due under Article 11 of the Nature Restoration Regulation.
The second is the compilation of validated occurrence records. The European Bee and Hoverfly Database (EBHD) assembles 4,341,684 bee and 1,036,181 hoverfly records with the taxonomic coverage reported above (Sentil et al., 2026; see Appendix 2 for a full description and for the limitations reported by its authors). Its completeness is unmatched, but its structure is not suited to estimating spatially resolved rates of change: the records are opportunistic and presence-only, with neither absences nor any measure of recording effort; data quality and quantity vary systematically between regions, highest in northern and western Europe and lowest in the south and east (Marshall et al., 2024); a substantial share is georeferenced to locality centroids rather than to collection sites, which is why 10 km is the finest resolution at which these records have been analysed within the Safeguard project itself.
Two consequences follow for the present assessment.
First, a rate of change estimated from these records would partly reflect the history of data production in each country, the timing of national digitisation, of taxonomic revisions, of targeted field campaigns in previously under-sampled regions, and of the recent expansion of photographic citizen-science

recording, rather than ecological change alone. Because that history is itself spatially structured, and inversely so with respect to the regions in which our results locate the largest impacts, a spatially explicit baseline derived on this basis would risk introducing a spurious spatial pattern into precisely the dimension the assessment is designed to resolve. Second, the core of this occurrence base derives from the BDFGM/Atlas Hymenoptera holdings and from the STEP project (Sentil et al., 2026), largely the same data lineage from which the bumblebee distribution models of Polce et al. (2018), underlying Prestele et al. (2021), and the ecological niche models of Ghisbain et al. (2024) are calibrated. A baseline trend estimated from these records would therefore not be independent of the climate channel already carried by the SDM ensemble, but would re-impose part of it on the same grid cells, compounding rather than resolving the overlap qualified above.

We therefore retain a spatially uniform degradation and treat its rate as a modelling assumption for scenario dimension. Sub-EU differentiation may become defensible in future once national Grassland Butterfly Indices are reported under Article 11, or once the EU Pollinator Monitoring Scheme established under Article 10 of Regulation (EU) delivers standardised, effort-controlled data for bees and hoverflies; both are developments of the coming years rather than currently available inputs.

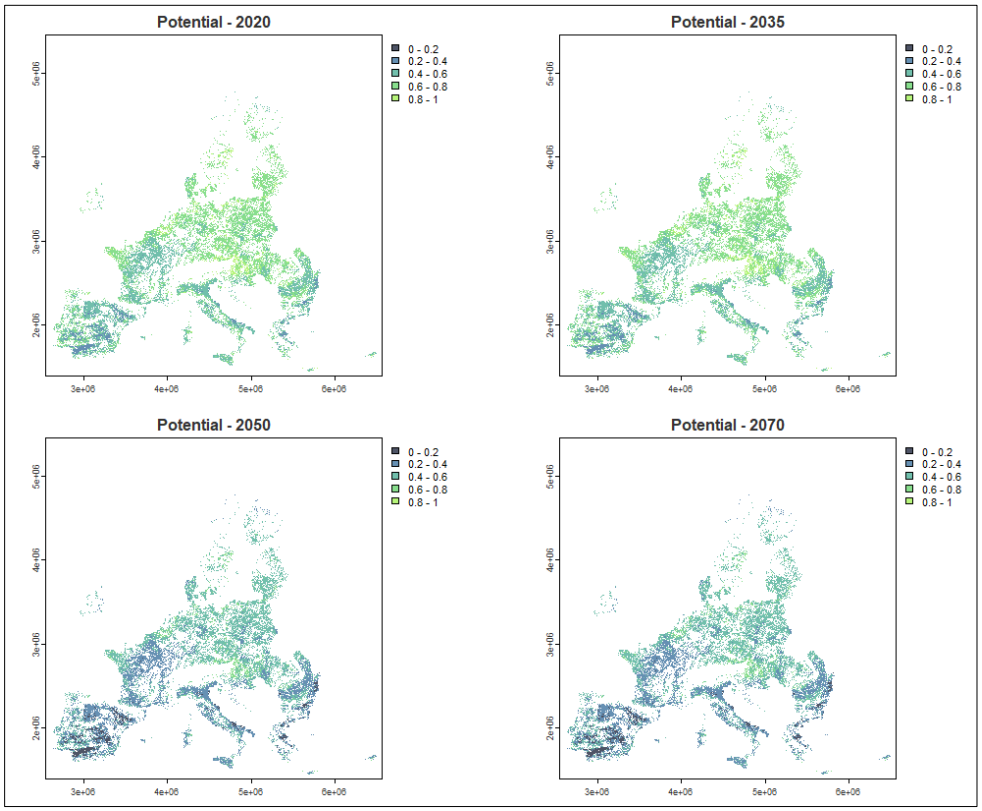


Figure A4: Projected pollination potential (0-1) under the baseline scenario (2020-2070) with no climate change.

Climate change scenarios

Projected declines in pollination potential due to changing climate are derived from an ensemble of state-of-the-art multi-model, multi-scenario species distribution models (SDMs) outputs for key pollinator taxa (wild bees, hoverflies, butterflies, beetles), based on the most recent and robust data available in literature. Data on habitat suitability from different external sources are imposed as shocks

on the baseline potential to detect the climate change and land use impacts on pollinator service under different scenarios and models' sensitivity.
This framework enables the assessment of multiple climate scenarios (RCP2.6, 4.5, 6.0, 7.0, 8.5) and socioeconomic pathways (SSP1, 2, 3, 5) to project climate-driven impacts on pollinators up to 2070. Data sources were selected to ensure full consistency with the KIP-INCA and Status and Trends of European Pollinators (STEP) frameworks. By mapping baseline pollination potential together with spatially explicit projections of climate change impacts on pollinators' habitat suitability, we estimate the expected future pollination potential and its percentage change relative to the baseline (2020–2070) under different climate change (RCPs) and land use (SSPs) scenarios, at a very high spatial resolution of 1 km × 1 km.

Scenario name	Land Use	RCP	Tot
PREST-CLIM	Constant	2.6 - 4.5 - 8.5	3
IIASA-CLIM	Constant	2.6 - 4.5 - 7.0 - 8.5	4
PREST-1	SSP1	2.6 - 4.5	2
PREST-3	SSP3	4.5 - 8.5	2
PREST-5	SSP5	8.5	1
IIASA-2	SSP2	2.6 - 4.5 - 7.0	3
GHSB-1	SSP1	2.6	1
GHSB-3	SSP3	6.0	1
GHSB-5	SSP5	8.5	1
All land-use / RCP combination considered			18

Figure A5: the pollination scenarios considered in this study as a combination of land use variables and climate radiative forcing (RCPs) used by each species distribution model (SDM)

To model **PREST-CLIM** and **PREST-1,3,5** scenarios, data from Prestele et al., 2021 are used to proxy the climate change impacts on pollinators affecting the baseline potential according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 1-4.5, SSP 3-4.5, SSP 3-8.5 and SSP 5-8.5 (+2.6, +4.5, +8.5 W m−2 of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium and high greenhouse gas emissions, respectively).
The choice was made because of data consistency with the INCA and MAES framework and Status and Trends of European Pollinators EU STEP project. Prestele et al., 2021 project the potential distribution of 47 European bumblebee species in 2050 and 2080 from existing European-scale distribution maps, based on a set of climate and land-use futures simulated through a regional integrated assessment model and consistent with the RCP–SSP scenario framework. Baseline bumblebee distributions were based on Species Distribution Model (Polce et al., 2018) and adjusted to be consistent with land-use data from the projections. Future climate and land-use data were taken from CMIP5 climate model simulations (Taylor et al., 2012) and the IAP2 (Harrison et al., 2019), respectively. They applied the SDMs to the time periods representative for the years 2050 and 2080 to obtain estimates of the future distributions of the 47 bumblebee species.
The potential distribution of 47 bumblebee species and SDM model coefficients were made available by Polce et al. (2018). The distribution maps indicate for each species (1) the probability of occurrence and (2) presence/absence at a spatial resolution of 10 x 10 km, based on a maximum entropy (MaxEnt)

modelling approach that establishes a functional relationship between species occurrence and 22 environmental predictor variables.

Various levels of climate change were considered through the scenario framework of the IPCC, combining representative concentration pathways (RCPs) with shared socioeconomic pathways (SSPs) (O'Neill et al., 2017; van Vuuren et al., 2011). Their findings show that direct climate impacts, although variable across species, dominate responses for most species, especially under high-end climate change scenarios (up to 99% range loss).

For our constant land use PREST-CLIM scenarios, we use data from the dynamic climate and constant land use (CLIM) experiment, where results are not determined by SSPs' related demand and land use change (Figure below), but only by different levels of radiative forcing (RCPs), so that in our modelling framework, future land use is held constant from FAO CROPGRIDS (Tang et al., 2024) datasets.

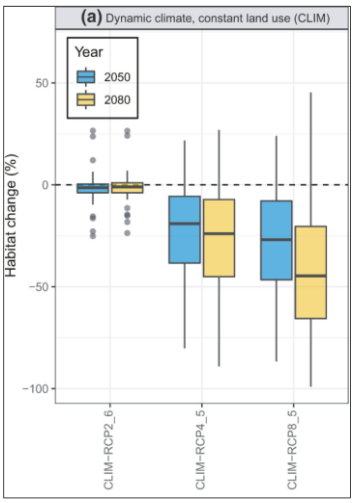


Figure A6: EU averaged habitat change for pollinators species under climate change and constant land use in 2050 and 2080. Source: Prestele et al., 2021

For the PREST-1,3,5 scenarios, we use data from dynamic climate and dynamic land use (COMB) experiment. Given a certain level of climate change (RCP2.6, RCP4.5, RCP8.5), including dynamic land-use change (COMB modelling experiment) has varying impacts on the species' response across land-use scenarios. In a world consistent with RCP2.6 climate in 2080, land use following the SSP1 storyline results in an average additional habitat change of −7.50%, which is largely driven by large losses of individual species. In contrast, SSP4 land-use changes almost offset average habitat losses from climate change (+2.09%). In the COMB-RCP2.6-SSP1 scenario 22 species lose additional habitat compared to the respective CLIM-RCP2.6 scenario, while 25 species gain habitat. In COMB-RCP2.6-SSP4 19 species lose habitat, while 28 gain habitat; but, more importantly, it is different species that are affected in the one or other direction. SSP1 land use leads to strong additional habitat losses for rare species (−31.21% for the lower quartile) while common species tend to benefit (+2.35% and +1.15% for the third and upper quartiles). SSP4 land use, in contrast, attenuates habitat losses for rare species (+11.37% for the lower quartile) and common species experience additional losses (−1.66% for third and −0.83% for the upper quartile). For 35 out of 47 species, the choice of land-use scenario has an opposite effect on the net change to suitable habitat area (i.e., habitat loss in one land-use scenario contrasts with habitat gain in the other) and for 24 species it has the potential to change the sign of the combined signal compared to the CLIM modelling experiment. Importantly, differences between land-use scenarios do also exist in RCP4.5 and RCP8.5 worlds in 2080, although the importance of land-use decreases with higher levels of climate change.

Overall, the data are spatially referenced and capture both increases and decreases in pollinator distributions as determined by projected regional climate conditions under the various SSP–RCP combinations. Consequently, some regions are expected to experience gains in pollination potential, while others will face substantial declines. This reflects the inherently local nature of climate impacts: evidence shows that many wild pollinator species are likely to shift towards northern ranges, whereas most bumblebee species are projected to undergo significant range contractions (Potts et al., 2015). By combining baseline pollination potential with spatially explicit projections of climate-induced changes in pollinator distributions, we estimate future pollination potential and its percentage change relative to the baseline (2020–2080) across multiple climate scenarios (RCP 2.6, 4.5, and 8.5), at a high spatial resolution of 1 km × 1 km.

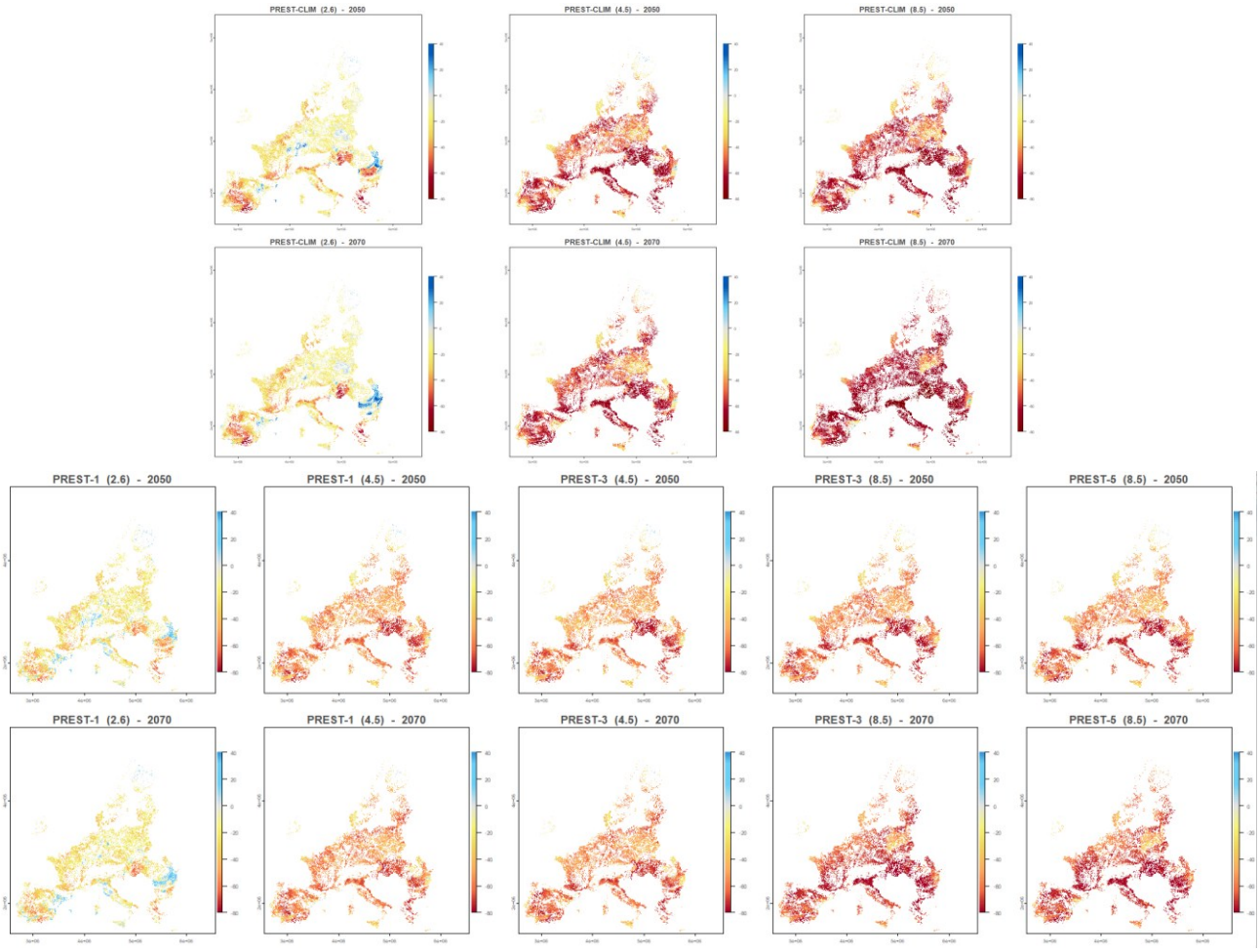


Figure A7: Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under PREST-CLIM and PREST-1,3,5 scenarios.

To model **GHSB-1,2,3** climate change scenarios, data from Ghisbain et al., 2024 are used to proxy the climate change impacts on pollinators affecting the baseline potential, according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to three distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 3-6.0 and SSP 5-8.5 (+2.6, +6.0 and +8.5 W m−2 of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium-high and high greenhouse gas emissions, respectively).

Authors analyse the past, present and future ecological suitability of the European continent for 46 bumblebee species on the basis of a set of harmonized climate, land use and human population data obtained through the Inter-Sectoral Impact Model Intercomparison Project Phase 2b (ISIMIP2b) for the last century, the present period and for upcoming years to 2080. The observational dataset comprises 401,046 unique georeferenced occurrence records of bumblebee presence split into 'past' (1901–1970 inclusively) and 'present-day' time periods (2000–2014 inclusively; Methods). To model the species–environment relationships between bumblebee occurrence records and the set of ISIMIP2b variables, they train ecological niche models on georeferenced data corresponding to each period using a boosted regression tree (BRT) approach. The resulting predictive probability is here interpreted as a measure of ecological suitability given local environmental conditions and varies between 0 (unsuitable environmental conditions) and 1 (highly suitable environmental conditions). In addition to the computation of commonly used area under the receiver operating characteristic curve (AUC) metric, they assessed the predictive performance of the ecological niche models by computing a prevalence-pseudoabsence-calibrated Sørensen's index. Using the same index, they then evaluated the predictive performance of our models by assessing the ability of ecological niche models trained on past occurrence and environmental data to predict present-day distributions when projected on present-day environmental data, as well as the ability of ecological niche models trained on present-day occurrence and environmental data to predict past distributions when projected on past environmental data.
In addition to training and mapping BRT models with the past and present-day occurrence data and environmental conditions, models trained on present-day data are further used to project the changes in ecological suitability maps in the next decades. For this purpose, they use environmental layers projected for three time windows (2021–2040, 2041–2060 and 2061–2080) according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to three distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 3-6.0 and SSP 5-8.5 (+2.6, +6.0 and +8.5 W m−2 of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium-high and high greenhouse gas emissions, respectively). They used three different global change trajectories instead of a single one to account for uncertainty surrounding future global changes, which are contingent on societal decisions.

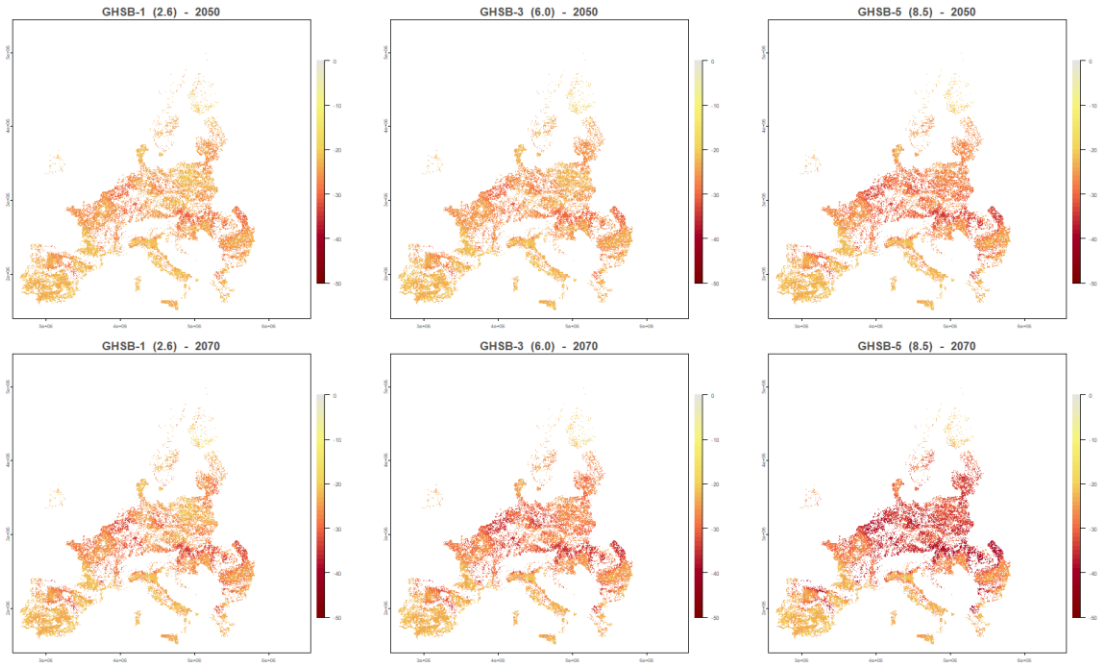


Figure A8: Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under GHSB-1,2,3 scenarios.

To model **IIASA-CLIM** and **IIASA-2** scenarios, we rely on IIASA external analysis on habitat suitability under climate and land-use scenarios from ACCREU project deliverable D2.4.[1] (Gerber G., Palazzo A. et al., 2025; Gerber, 2026 a; b), to proxy climate change impacts on pollinators imposed on our baseline potential. They utilized individual species distribution models (SDMs) for key pollinator groups Lepidoptera (Butterflies) and Hymenoptera (Bees), representing, along with Coleoptera and Diptera (not modelled due to data limitations), the dominant taxa providing pollination services to crops in Europe.

Pollinator Group	Species count	% of Total Pollinators	Represented Genus (Species count per genus)
Hymenoptera (Bees)	96	62.75	*Andrena (10); Bombus (11); Chelostoma (1); Colletes (2); Dufourea (1); Epeolus (2); Halictus (2); Hoplitis (3); Hylaeus (2); Lasioglossum (2); Macropis (1); Megachile (3); Nomads (12); Osmia (1); Panurgus (2); Sphecodes (1); Stelis (1)*
Lepidoptera (Butterflies)	57	37.25	*Anthocharis (1); Apatura (1); Aricia (1); Arytrura (1); Cacyreus (1); Clossiana (1); Coenonympha (6); Colias (1); Cupido (1); Danaus (1); Erannis (1); Erebia (20); Eriogaster (1); Euchloe (2); Euphydryas (2); Gegenes (1); Glyphipterix (1); Hipparchia (7); Laeosopis (1); Lampides (1); Lasiommata (1); Leptidea (1); Leptotes (1); Lignyoptera (1); Lopinga (1); Lycaena (3); Lysandra (1); Melanargia (2); Melitaea (4); Nymphalis (2); Oeneis (1); Papilio (2); Parnassius (2); Pelopidas (1); Pieris (1); Polyommatus (4); Pontia (1); Proserpinus (1); Pseudophilotes (2); Pyrgus (6); Vanessa (3); Xestia (1); Xylomoia (1); Zerynthia (1)*
Total	**153**	**100 %**	

Figure A9: Summary of wild insect pollinator groups (Hymenoptera, Lepidoptera) considered in the IIASA scenarios. Source: ACCREU (EU) Deliverable D2.4 (Gerber G., Palazzo A. et al., 2025)

The SDMs provide species-specific habitat availability indices, represented as a fraction of suitable habitat per 10 km 2 grid cell across Europe for different climate and adaptation scenarios from 2015 to 2095 in 10-year intervals. From these, they calculated species-weighted habitat availability across all pollinator species for each 10 km 2 grid cell, providing a measure of overall pollinator potential (0-1). The analysis captured the combined impact of future climate and land use on individual species distributions by coupling a species distribution modelling approach with the ibis.iSDM modelling tool (Jung, 2023a) and refinement of suitable habitat. The IIASA-CLIM scenarios are based on the 'climate_only_mean_suitability' experiment which relies on the 2015/2025 baseline land use data and future modelled GCMs projections from ISIMIP3b climate forcing data. Full data and method can be found in Gerber, 2026 (b). European pollinator species distribution models and pollination supply metrics under climate change (Version 1.0) DOI: 10.5281/zenodo.22751032). In the dataset used for the IIASA-2 scenarios, land use projections from the downscaled GLOBIOM land use maps were also used. Full data and method can be found in Gerber, 2026 (a). Dataset ACCREU D2.4: European crop pollination supply, demand, and flow under climate and land-use change (Version 1.0) DOI: 10.5281/zenodo.22743969).
The resulting projections capture the climatic niche (unit 0-1) of a species for each SSP and adaptation. For each species, they calculated within the current and future climatic niche the amount of suitable

[1] ACCREU (EU), 2025: Deliverable D2.4 (Impacts on Ecosystems & Biodiversity). GA number: 101081358; Funding type: HORIZON-CL5-2022-D1-07-two-stage.

habitat, integrating information on downscaled land-use change for each SSP-RCP scenario combination and species-specific habitat preferences, using the ibis.insights R package (Jung, 2023b).

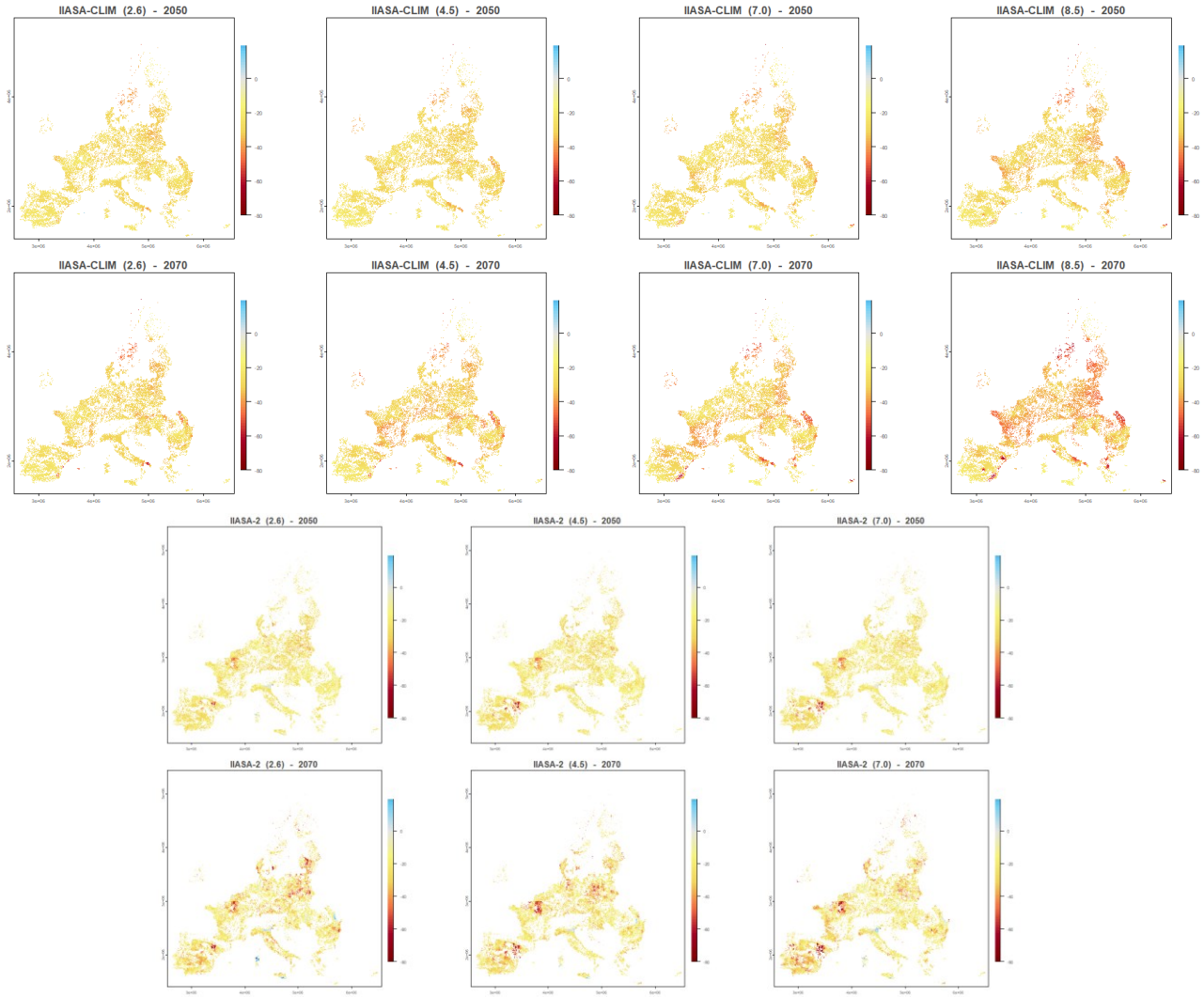


Figure A10: Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under IIASA-CLIM and IIASA-2 scenarios.

Aggregation of pollination shocks to GTAP sectors

The projected gridded percentage changes in yield for each crop and climate scenario are mapped and aggregated at NUTS levels (0 to 3) and then used as land-productivity shocks in the macroeconomic model. Macroeconomic impacts are then estimated using a regionalized version of the ICES Computable General Equilibrium (CGE) model (Standardi et al. 2023; van der Wijst et al. 2023) based on GTAP11 database and parameters (Aguiar et al., 2022), which provides detail for 122 regions down to NUTS-2 level across the EU27, while the rest of the world is represented as a macro-regional aggregate. The theoretical structure of the CGE model is based on the GTAP model (Hertel, 1997). Representative optimizing agents (households, firms) maximise an objective function subject to a specific constraint. Households maximise the utility flow from consumption subject to a budget constraint. Firms maximise profits subject to a technological constraint. Labour and capital are fully employed and mobile across sectors but not between regions. Investment is saving driven, and trade

between subnational regions follows the Armington assumption (imperfect substitutability between domestic and foreign product). The model is recursive-dynamic, this implies that agents have myopic expectations (no perfect foresight) and the investment computed in the previous time period is added to the depreciated capital in the subsequent time step.

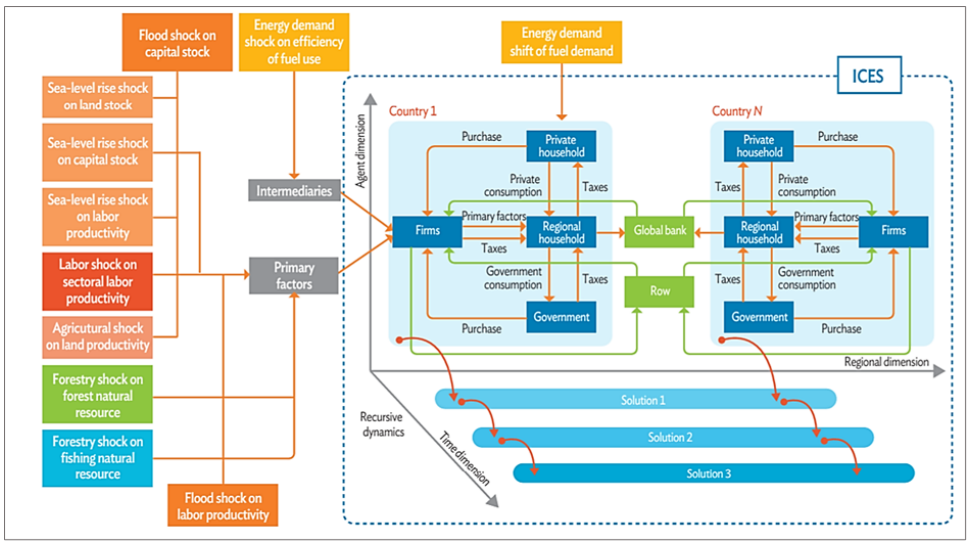


Figure A11: ICES model and channels of implementation of climate change-related shocks. Source: Campagnolo et al., 2026

To perform the macroeconomic assessment of our exercise, the GTAP11 database agricultural sub-sectors entering the CGE model have been aggregated in crop categories as follows:

	GTAP 11	GTAP 11 description	Sect aggr	Paper description
1	pdr	Paddy rice	gro	Cereals, wheat, rice
2	wht	Wheat	gro	Cereals, wheat, rice
3	gro	Cereal grains nec	gro	Cereals, wheat, rice
4	v_f	Vegetables, fruit, nuts	v_f	Vegetables, fruit, nuts
5	osd	Oil seeds	osd	Oil seeds
6	c_b	Sugar cane, sugar beet	c_b	Sugar cane, sugar beet
7	pfb	Plant-based fibers	ocr	Other crops, fibers
8	ocr	Crops nec	ocr	Other crops, fibers

Since the resulting agricultural sector in the CGE is relatively aggregated, we averaged the projected yield percentage changes based on the relative weight of each crop cultivated in each region, aligning them with the crop categories used in the model. This approach is also convenient because the GTAP crop categories generally correspond to similar levels of pollination dependence; for example, Vegetables and fruits have, on average, high dependence, Oil seeds medium, while Cereals, Wheat and Rice low to negligible dependence. To do this, we compute a weighted average in which the weight of the specific crop in each grid cell within a NUTS region is proportional to the number of hectares cultivated in that cell. These results are then further aggregated to match the crop categories used in the economic model, taking into account the relative contribution of each crop within the NUTS region of interest. With the exception of the CLIM scenarios, which are based on a constant 2020 land use exercise, we incorporate projected crop and crop category-specific land-use changes from Goebel and Britz (2025), to account for future land-use changes and crop expansion under different SSPs (Riahi et al., 2017, van Vuuren et al., 2014) at the highest resolution available. The SSPs-consistent land-use projections from Goebel and Britz (2025) are chosen to ensure full consistency with our modelling

framework. They also rely on a Computable General Equilibrium platform built on the GTAP v11 Power Database (Aguiar et al., 2022). It represents the global economy in 2017 and includes LULC satellite information, with regional land-use extents adjusted using additional datasets such as the 2018 European CORINE land-cover (LC) data (Corine, 2018). Cropland and yield developments are calibrated to projections from the Food and Agriculture Organization of the United Nations (FAO, 2018), thereby maintaining full alignment across all components of our modelling chain. Land-use change information is provided at the NUTS-2 level for eight land-cover types (e.g., cropland, forests, pastureland, shrubland) and for 65 land-use categories corresponding to individual crops and crop aggregates, which are matched to the FAO CROPGRIDS crop categories used in this study.

To ensure consistency between the pollination projections, future yield changes and the model factor endowments, we also impose projected SSP-consistent agricultural land changes for the NUTS EU regions, from Goebel and Britz (2025), both in the baseline runs and in the impact scenarios under different SSPs.

Mapping the 122 ICES EU Regions and the Eurostat NUTS code 2021

Country	Nuts Code	Level	ICES Name	Country	Nuts Code	Level	ICES Name	Country	Nuts Code	Level	ICES Name
Austria	AT1	1	EastAustria	Germany	DEF	1	SchHol	Portugal	PT18	2	Alentejo
Austria	AT2	1	SouthAustria	Germany	DEG	1	Turingia	Romania	RO1	1	RomanOne
Austria	AT3	1	WestAustria	Greece	EL3	1	Attiki	Romania	RO2	1	RomanTwo
Belgium	BE1	1	Flanders	Greece	EL4	1	Nisia	Romania	RO3	1	RomanThree
Belgium	BE3	1	Wallonia	Greece	EL5	1	Voreia	Romania	RO4	1	RomanFour
France	FR10	2	IleFrance	Greece	EL6	1	Kentriki	Spain	ES11	2	Galicia
France	FRB0	2	Centre	Italy	ITC1	2	Piemonte	Spain	ES12	2	Asturias
France	FRC1	2	Bourgogne	Italy	ITC2	2	ValAosta	Spain	ES13	2	Cantabria
France	FRC2	2	FranComte	Italy	ITC3	2	Liguria	Spain	ES21	2	PaisVasco
France	FRD1	2	BasseNorm	Italy	ITC4	2	Lombardia	Spain	ES22	2	Navarra
France	FRD2	2	HautNorm	Italy	ITH1	2	TrentAdige	Spain	ES23	2	LaRioja
France	FRE1	2	NordPCalais	Italy	ITH3	2	Veneto	Spain	ES24	2	Aragon
France	FRE2	2	Picardie	Italy	ITH4	2	FriuliGiulia	Spain	ES30	2	Madrid
France	FRF1	2	Alsace	Italy	ITH5	2	EmiRom	Spain	ES41	2	CastLeon
France	FRF2	2	ChamArde	Italy	ITI1	2	Toscana	Spain	ES42	2	CastMancha
France	FRF3	2	Lorraine	Italy	ITI2	2	Umbria	Spain	ES43	2	Extremadura
France	FRG0	2	PaysLoire	Italy	ITI3	2	Marche	Spain	ES51	2	Cataluna
France	FRH0	2	Bretagne	Italy	ITI4	2	Lazio	Spain	ES52	2	Valencia
France	FRI1	2	Aquitaine	Italy	ITF1	2	Abruzzo	Spain	ES53	2	Balears
France	FRI2	2	Limousin	Italy	ITF2	2	Molise	Spain	ES61	2	Andalucia
France	FRI3	2	PoitouChar	Italy	ITF3	2	Campania	Spain	ES62	2	Murcia
France	FRJ1	2	LangRouss	Italy	ITF4	2	Puglia	Spain	ES70	2	Canarias
France	FRJ2	2	MidiPyren	Italy	ITF5	2	Basilicata	Sweden	SE1	1	EastSweden
France	FRK1	2	Auvergne	Italy	ITF6	2	Calabria	Sweden	SE2	1	SouthSweden
France	FRK2	2	RhoneAlp	Italy	ITG1	2	Sicilia	Sweden	SE3	1	NorthSweden
France	FRL0	2	Provence	Italy	ITG2	2	Sardegna	Bulgaria	BG	0	bgr
France	FRM0	2	Corse	Netherlands	NL1	1	NorthNether	Croatia	HR	0	hrv
Germany	DE1	1	BadenWur	Netherlands	NL2	1	EastNether	Cyprus	CY	0	cyp
Germany	DE2	1	Bavaria	Netherlands	NL3	1	WestNether	Czech Republic	CZ	0	cze
Germany	DE3	1	Berlin	Netherlands	NL4	1	SouthNether	Denmark	DK	0	dnk
Germany	DE4	1	Branden	Poland	PL2	1	SouthPol	Estonia	EE	0	est
Germany	DE5	1	Bremen	Poland	PL4	1	NorWestPol	Finland	FI	0	fin
Germany	DE6	1	Hamburg	Poland	PL5	1	SouWestPol	Hungary	HU	0	hun
Germany	DE7	1	Hessen	Poland	PL6	1	NorthPol	Ireland	IE	0	irl
Germany	DE8	1	MeklenVor	Poland	PL7	1	CentrePol	Latvia	LV	0	lva
Germany	DE9	1	LowSaxony	Poland	PL8	1	EastPol	Lituania	LT	0	ltu
Germany	DEA	1	NorRenoWes	Poland	PL9	1	Masovia	Luxembourg	LU	0	lux
Germany	DEB	1	RenoPala	Portugal	PT11	2	Norte	Malta	MT	0	mlt
Germany	DEC	1	Saarland	Portugal	PT15	2	Algarve	Slovakia	SK	0	svk
Germany	DED	1	Saxony	Portugal	PT16	2	Centro	Slovenia	SI	0	svn
Germany	DEE	1	SaxonyAnh	Portugal	PT17	2	Lisboa				

Figure A12: Mapping between the 122 ICES EU regions used in this study, and the correspondent 2021 Eurostat NUTS code.

The table below shows the aggregation matches performed for the economic impact projections between:
(i) the crops and crop categories (Crop) from FAO CROPGRIDS (Tang et al., 2024)
(ii) the corresponding sector (GTAP11 sec) in the CGE model
(iii) the dependency rate to wild pollination (Dep. Rate)
(iv) the land use categories (Land-Use) from the SSPs projections from Goebel and Britz (2025).

	Crop	GTAP 11 sec		Dep. Rate	Land Use
1	alfalfa	ocr	Crops nec	0.95	ocro
2	almond	v_f	Vegetables, fruit, nuts	0.65	ofru
3	aniseetc	ocr	Crops nec	0.05	ocro
4	apple	v_f	Vegetables, fruit, nuts	0.65	appl
5	apricot	v_f	Vegetables, fruit, nuts	0.65	ofru
6	artichoke	v_f	Vegetables, fruit, nuts	0.25	oveg
7	asparagus	v_f	Vegetables, fruit, nuts	0.05	oveg
8	avocado	v_f	Vegetables, fruit, nuts	0.65	oveg
9	banana	v_f	Vegetables, fruit, nuts	0.05	banp
10	bean	ocr	Crops nec	0.05	leg
11	beetfor	ocr	Crops nec	0.05	ocro
12	berrynes	v_f	Vegetables, fruit, nuts	0.50	ofru
13	blueberry	v_f	Vegetables, fruit, nuts	0.65	ofru
14	broadbean	v_f	Vegetables, fruit, nuts	0.25	ofru
15	buckwheat	ocr	Crops nec	0.65	ocro
16	cabbage	v_f	Vegetables, fruit, nuts	0.05	oveg
17	cabbagefor	v_f	Vegetables, fruit, nuts	0.05	oveg
18	carob	v_f	Vegetables, fruit, nuts	0.25	oveg
19	castor	osd	Oil seeds	0.05	oso
20	cauliflower	v_f	Vegetables, fruit, nuts	0.05	oveg
21	cherry	v_f	Vegetables, fruit, nuts	0.65	ofru
22	chestnut	v_f	Vegetables, fruit, nuts	0.25	ofru
23	chickpea	ocr	Crops nec	0.25	leg
24	chicory	v_f	Vegetables, fruit, nuts	0.25	oveg
25	chilleetc	ocr	Crops nec	0.25	leg
26	citrusnes	v_f	Vegetables, fruit, nuts	0.65	citr
27	cotton	ocr	Crops nec	0.25	pfb
28	cowpea	ocr	Crops nec	0.05	leg
29	cranberry	v_f	Vegetables, fruit, nuts	0.65	ofru
30	cucumberetc	v_f	Vegetables, fruit, nuts	0.65	oveg
31	currant	v_f	Vegetables, fruit, nuts	0.05	ofru
32	date	v_f	Vegetables, fruit, nuts	0.05	ofru
33	eggplant	v_f	Vegetables, fruit, nuts	0.25	oveg
34	fig	v_f	Vegetables, fruit, nuts	0.65	ofru
35	flax	v_f	Vegetables, fruit, nuts	0.25	oveg
36	fornes	v_f	Vegetables, fruit, nuts	0.05	oveg
37	fruitnes	v_f	Vegetables, fruit, nuts	0.65	ofru
38	gooseberry	v_f	Vegetables, fruit, nuts	0.25	ofru
39	grape	v_f	Vegetables, fruit, nuts	0.05	grap
40	grapefruitetc	v_f	Vegetables, fruit, nuts	0.05	grap
41	greenbean	v_f	Vegetables, fruit, nuts	0.05	oveg
42	greenbean	ocr	Crops nec	0.25	leg
43	greencorn	gro	Cereal grains nec	0.05	maiz
44	greenonion	v_f	Vegetables, fruit, nuts	0.05	oveg
45	groundnut	osd	Oil seeds	0.05	oso
46	kiwi	v_f	Vegetables, fruit, nuts	0.95	ofru
47	legumenes	ocr	Crops nec	0.25	leg
48	lemonlime	v_f	Vegetables, fruit, nuts	0.05	citr
49	lettuce	v_f	Vegetables, fruit, nuts	0.05	oveg
50	linseed	osd	Oil seeds	0.25	oso
51	maize	gro	Cereal grains nec	0.05	maiz
52	maizefor	gro	Cereal grains nec	0.05	maiz
53	melonetc	v_f	Vegetables, fruit, nuts	0.65	ofru
54	melonseed	v_f	Vegetables, fruit, nuts	0.65	ofru
55	mixedgrain	gro	Cereal grains nec	0.05	ocer
56	mixedgrass	ocr	Crops nec	0.05	pfb
57	mustard	ocr	Crops nec	0.25	leg
58	nutnes	v_f	Vegetables, fruit, nuts	0.65	ofru
59	oilseedfor	osd	Oil seeds	0.25	oso
60	oilseednes	osd	Oil seeds	0.25	oso
61	okra	v_f	Vegetables, fruit, nuts	0.25	oveg
62	olive	osd	Oil seeds	0.05	olv
63	onion	v_f	Vegetables, fruit, nuts	0.05	oveg
64	orange	v_f	Vegetables, fruit, nuts	0.25	citr
65	peachetc	v_f	Vegetables, fruit, nuts	0.65	ofru
66	pear	v_f	Vegetables, fruit, nuts	0.65	ofru
67	peppermint	ocr	Crops nec	0.25	ocro
68	persimmon	v_f	Vegetables, fruit, nuts	0.05	ofru
69	pimento	v_f	Vegetables, fruit, nuts	0.25	ofru
70	pineapple	v_f	Vegetables, fruit, nuts	0.25	ofru
71	plum	v_f	Vegetables, fruit, nuts	0.65	ofru
72	poppy	ocr	Crops nec	0.25	leg
73	potato	v_f	Vegetables, fruit, nuts	0.05	pota
74	pulsenes	ocr	Crops nec	0.25	leg
75	pumpkinetc	v_f	Vegetables, fruit, nuts	0.95	oveg
76	quince	v_f	Vegetables, fruit, nuts	0.25	oveg
77	rapeseed	osd	Oil seeds	0.25	rape
78	rasberry	v_f	Vegetables, fruit, nuts	0.65	ofru
79	safflower	osd	Oil seeds	0.05	oso
80	sesame	osd	Oil seeds	0.25	oso
81	sisal	osd	Oil seeds	0.25	oso
82	sourcherry	v_f	Vegetables, fruit, nuts	0.65	ofru
83	soybean	ocr	Crops nec	0.25	soy
84	spinach	v_f	Vegetables, fruit, nuts	0.05	oveg
85	stonefruitnes	v_f	Vegetables, fruit, nuts	0.65	ofru
86	strawberry	v_f	Vegetables, fruit, nuts	0.65	ofru
87	stringbean	ocr	Crops nec	0.25	leg
88	sugarbeet	c_b	Sugar cane, sugar beet	0.05	c_b
89	sugarcane	c_b	Sugar cane, sugar beet	0.05	c_b
90	sunflower	osd	Oil seeds	0.45	oso
91	swedefor	v_f	Vegetables, fruit, nuts	0.05	pota
92	sweetpotato	v_f	Vegetables, fruit, nuts	0.05	pota
93	tangetc	v_f	Vegetables, fruit, nuts	0.05	oveg
94	taro	ocr	Crops nec	0.25	ocro
95	tea	ocr	Crops nec	0.25	ocro
96	tobacco	ocr	Crops nec	0.25	ocro
97	tomato	v_f	Vegetables, fruit, nuts	0.25	toma
98	tropicalnes	v_f	Vegetables, fruit, nuts	0.25	ofru
99	turnipfor	ocr	Crops nec	0.65	ocro
100	vegetablenes	v_f	Vegetables, fruit, nuts	0.65	oveg
101	vegfor	v_f	Vegetables, fruit, nuts	0.25	oveg
102	watermelon	v_f	Vegetables, fruit, nuts	0.95	ofru
103	yam	ocr	Crops nec	0.25	ocro

Figure A13: Mapping between FAO crop categories, GTAP11 sectors, land use projections and dependency rates.

The dependency rates reported here measure the average impact on production of pollination loss, i.e. the fractional reduction in crop commercial yield associated with a loss of animal pollination (Klein et al., 2007; Gallai et al., 2009; Bugin et al., 2022; Siopa et al., 2024; Ryan et al., 2023). For example, a pollination dependency of 0.05, means that a 100% decrease in pollination results in a 5% decrease in commercial yield. The impact on production for a specific crop in a given area is therefore determined

by the projected change in pollination service in the area where the crop is cultivated and by that crop's dependency on animal pollination. Approximation of long term effects across 129 crop categories is unavoidable at this resolution and time horizon, while commonly applied methods have been shown to underestimate pollinator dependence values (Siopa et al., 2024). Dependency rates were taken from literature using multiple sources, also accounting for indirect effects on crop quality when this significantly influences marketable yield (Klein, 2007; Aizen et al., 2009; Bauer, 2016; Perrot et al., 2019; Bugin et al., 2022; Munir et al., 2020; Layek et al., 2023). Crops whose marketable product is vegetative (roots, tubers, bulbs, leaves, stems, fibre or forage biomass) are assigned the lowest classes or zero, since pollination does not affect the yield harvested in a given season: of 55 such crop layers, 52 carry a rate of 0.25 or below and 17 are set to zero, including clover, vetch and lupine. Alfalfa is a declared exception, retaining the rate assigned to its seed production (0.95): stands are periodically re-established from certified seed that is itself pollinator-essential, so over a fifty-year horizon the dependency operates through the seed supply rather than within the season, and the rate is an upper bound of that channel. For several crops the rates adopted here are lower than those reported by recent crop-specific assessments: animal pollination has been estimated to account for close to 60% of citrus yield per hectare (Monasterolo et al., 2024), against the 0.05-0.25 used here. Our estimates of pollination-driven yield loss are therefore conservative for these crops.

Appendix 2. Pollination current status and projected trends

Current Status of European Pollinators

The IUCN European Red List of Bees (Nieto et al., 2014) shows that 9% of all bee species are threatened, with negative population trends for more than 45% of species, while 26% of bumblebees are classified as threatened. This primarily reflects the loss and fragmentation of semi-natural grasslands, and pressures from intensive agriculture and pesticides. Overall, 57% of all species were data deficient and could not be assessed at the EU level.

The 2026 re-assessment, produced under the DG ENV PULSE project, substantially worsens this picture: nearly 100 additional wild bee species are now classified as threatened and more than 20% of species in groups such as bumblebees and cellophane bees face extinction risk, with endemism and the largest residual data gaps both concentrated in southern Europe, particularly Spain and Greece (Michez et al., 2026). The two other pollinator groups have since been assessed at the same continental scale: 37% of the 890 European hoverfly species are threatened, with 244 species affected by habitat degradation and shifts driven by climate change and the associated increase in fire frequency (Vujić et al., 2022), while the number of threatened European butterfly species increased by 76% over the last decade, 28.3% of species now being threatened or Near Threatened in Europe and 30.9% in the EU27, and 52% of threatened species being affected by climate change (Van Swaay et al., 2025a).

The underlying distributional evidence base has also been consolidated: national checklists now cover some 3,000 European bee and hoverfly species (Reverté et al., 2023) and a validated occurrence database compiles more than 4.34 million bee and 1.04 million hoverfly records covering 97% of both European faunas on a 10 km resolution (Sentil et al., 2026).

Since the 1950s, wild bees in the Netherlands, Belgium and Great Britain have generally declined in diversity and occurrence (Biesmeijer et al., 2006; Carvalheiro et al., 2013). There is evidence of parallel declines in bee species richness in many areas, along with declines in plant species that are reliant on insect pollinators in Great Britain or bee pollinators in the Netherlands, when compared to declines of wind- or water-pollinated plant species. Of the 137 UK wild bee species analysed between 1980 and 2016, 37% declined in occupancy and 20% increased, while the average trend across the species was a 25% decline. The declines were greatest between 2006 and 2013, and the average trend across species has since stabilised (Powney et al., 2019). Since 1909, 20 bee and wasp species have become extinct in Britain (Ollerton et al., 2014).

Although trends cannot currently be extrapolated for many pollinator groups or at regional scales (Potts et al., 2020), strong evidence shows that declines in bees and hoverflies reported in the European Red Lists and national studies are expected to continue across most EU agricultural land. Therefore, the EU Grassland Butterfly Index (https://data.europa.eu/data/datasets/) is commonly used as a proxy for overall pollinator decline in Europe.

Butterflies are extensively monitored across Europe through the European Butterfly Monitoring Schemes (https://butterfly-monitoring.net/). They occupy similar habitats and are highly sensitive to the same environmental pressures as many other pollinator groups, and their trends therefore mirror the responses of other pollinators for which comparable large-scale data are lacking (Van Swaay et al., 2022; 2025b). The number of threatened European butterfly species has increased by 76\% over the last decade, and more than 20% of bumblebee species, nearly 40% of hoverfly species, and 9\% of bees are threatened with extinction (Van Swaay et al., 2025b; eea2025).

The European Grassland Butterfly Indicator integrates the population trends of 17 butterfly species monitored across the EU and is calculated from data from all 27 EU Member States, starting from 1990 until 2023. The 1990-2023 index shows a linear decline of 32% in the EU27 and 36% in Europe in the period 2010-2020. The 2025 update of the European Butterfly Monitoring Schemes eBMS database (https://butterfly-monitoring.net/ebms-database) shows a 47% decline in the 1990-2024 period, equivalent to a -1.9% annual decrease (Van Swaay et al., 2025b).

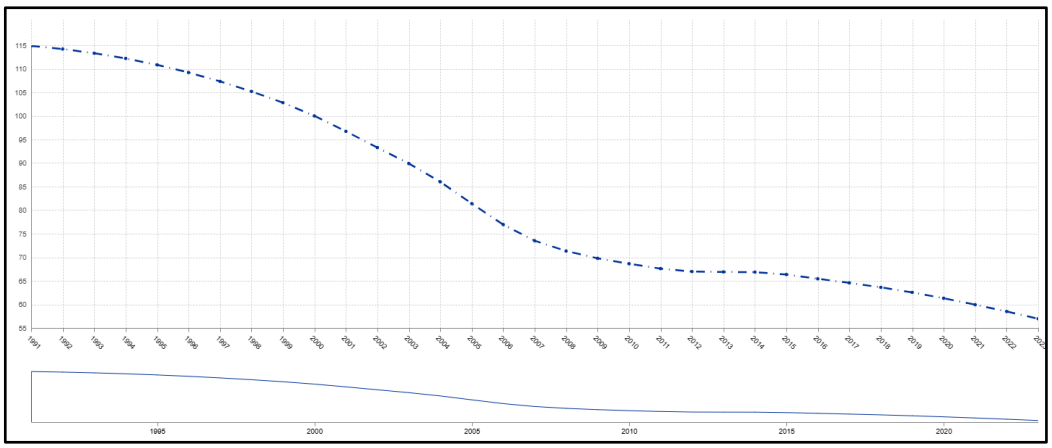


Figure A14: EU Grassland Butterfly Index 1990/2023 - EU aggregate from Eurostat data

The EU (2023) Impact Assessment Study to support the development of legally binding EU nature restoration targets assessed expected pollinator abundance trends if no additional action is taken. Extrapolations show that should the historical trend continue, the grassland butterfly indicator would likely decline by -23% by 2030, and by -48% by 2050, compared to 2017 values, meaning an average annual decline of -1.23%. This figure is consistent with the EU estimate of -1.5% annual degradation trend of agro-ecosystems (areas requiring restoration or conservation) (EU 2023, a, b). Both figures are

EU27 aggregates; the harmonised indicator is not published at national level, and national Grassland Butterfly Indices will become available only through the National Restoration Plans due under Article 11 of the Nature Restoration Regulation from September 2026 onwards.
Reversing the decline of pollinators is an objective of the EU Biodiversity Strategy for 2030 (https://environment.ec.europa.eu/strategy/biodiversity-strategy-2030_en) and the EU Pollinators Initiative (https://environment.ec.europa.eu/topics/nature-and-biodiversity/pollinators_en).
The EU 2024 Regulation on Nature Restoration (https://green-forum.ec.europa.eu/nature-and-biodiversity/nature-restoration-regulation-implementation_en) paves the way for a broad range of ecosystems to be restored. The Grassland Butterfly Index is included among three indicators in agricultural ecosystems from which Member States must select at least two, with the obligation to put measures in place which aim to achieve an increasing trend in the chosen indicators by 2030 and thereafter until satisfactory levels are achieved.

Climate Change Impacts on Pollinators

Habitat degradation and climate change are globally acting as pivotal drivers of wildlife collapse, with mounting evidence that this erosion of biodiversity will accelerate in the following decades (Ghisbain et al., 2024; IPBES, 2019). Moreover, although climate change impacts to date are not likely to be among the greatest drivers, they are expected to increase (IPCC, 2020), together with indirect drivers such as increased irrigation and production of bioenergy. Projections of common farmland bird and grassland butterfly indicators suggest that these groups will continue to decline into the future, likely driven by agricultural practices and land use. Accordingly, conservation and restoration measures are likely to increase pollinator species diversity and abundance (Olmeda et al., 2019).
Climate change will play an important role in future trends, disrupting seasonal temperatures and precipitation patterns. Rising temperatures are expected to increasingly impact pollinators, influencing their geographic ranges, population sizes, and seasonal activity patterns (Hoegh-Guldberg et al., 2019; Rahimi et al., 2024; 2025), possibly leading to extinctions of some species at the southern and northern edges of ranges and in alpine species, whilst driving the expansion of other species to colonise new areas (EU, 2023).
Bumblebees are particularly vulnerable as cold-adapted species. The average temperature requirements of all bumblebee species do not reflect the broad climatic range of the EU but are instead concentrated in intermediate to cold conditions (Rasmont et al., 2015). For European bumblebees, this range extends from -1.6°C (Bombus hyperboreus) to 10.4°C (B. ruderatus), with a median of 7.0°C. This value, known as the Community Temperature Index (CTI) (Devictor et al., 2012), is commonly used to assess community-level changes in response to climate change.
Attribution evidence at continental scale is now available for butterflies: analysing more than 8,400 populations of 145 species across 869 sites in 12 European countries over 1976–2021, Colom et al. (2025) find that climate warming is associated with population declines that are more pronounced in urban than in rural sites, that species with lower reproductive rates are most negatively affected, and that increasing precipitation benefits rural but harms urban populations. Consistently, threatened European butterflies are spatially concentrated in areas subject to strong climatic change and high atmospheric nitrogen deposition (Rashid et al., 2023), and the drivers of grassland butterfly decline differ regionally, with agricultural intensification, nitrogen deposition and pesticide drift dominating in north-western Europe and grassland abandonment together with climate change dominating in northern, eastern and southern Europe and in the Alps (Warren et al., 2021; Van Swaay et al., 2025b).

Evidence from Species Distribution Models

The overall impact of climate change on many pollinator species is difficult to assess, as for most species we often lack the long-term data and mechanistic evidence needed to identify climate-driven declines and predict future trends (Kazenel et al., 2024).

The EU Status and Trends of European Pollinators (STEP) project (Potts et al., 2015) found that the vast majority of bumblebees (up to 46 species in 2050 and up to 52 species in 2100) will suffer from range contractions under high-end emission scenarios (3°C and 4.7°C). Results of two large-scale studies projected the potential future distribution in Europe for butterflies by 2080 (Settele et al., 2008) and bumblebees by 2100 (Rasmont et al., 2015). Under the extreme (+5.6°C) scenario, 16% of the modelled European bumblebees and 24% of the analysed butterflies are projected to be at extremely high climatic risk, 29% and 16% will be at very high risk, 46% and 30% are at high risk, 5% and 24% are at risk, and only 2% and 6% are at low risk. Only a limited number of species are projected to benefit from climate change under a full dispersal assumption.

Across these assessments the binding constraint is the availability of standardised long-term data rather than modelling capacity, and the same constraint bounds the ensemble assembled here. The projections we combine resolve 153 species of Hymenoptera and Lepidoptera, 46 to 47 Bombus species in the Ghisbain and Prestele families, and 96 bee and 57 butterfly species in the IIASA family, against European checklists of 2,138 bee and 913 hoverfly species (Reverté et al., 2023; Sentil et al., 2026). Coverage is therefore partial and taxonomically non-random, weighted towards bumblebees, with Diptera and Coleoptera unrepresented for lack of data despite their documented contribution to crop pollination (Rader et al., 2016; Wardhaugh, 2015). Our yield shocks should accordingly be read as the response of the modelled subset rather than of the full pollinator assemblage; since bumblebees are cold-adapted and therefore among the taxa most exposed to warming (Rasmont et al., 2015), that subset is plausibly more climate-sensitive than the assemblage as a whole, which reinforces the reading of the ensemble as informative on the direction and spatial pattern of change rather than on its absolute magnitude.

Recent Multi-Species Projections:

Ghisbain et al. (2024) identified future large-scale range contractions and species extirpations under all future climate and land use change scenarios for 46 bumblebee species (Apidae, genus Bombus), for which long-term observation data, literature on biogeography and life history are available at the European spatial scale. Both environmental variables and occurrence data are used to generate ecological suitability through species-specific ecological niche modelling analyses that are then combined to compute an Ecological Suitability Index (ESI) and its evolution through time. Results show that temperature and precipitation have strong negative correlations with ecological suitability differences, suggesting that warming and altered rainfall patterns are major contributors to habitat loss, especially for cold-adapted species. Around 38-76% of studied European bumblebee species currently classified as 'Least Concern' are projected to undergo losses of at least 30% of ecologically suitable territory by 2061-2080 compared to 2000-2014.

Prestele et al. (2021) assessed the potential distribution of 47 European bumblebee species in 2050 and 2080 from existing European-scale distribution maps, based on a set of climate and land-use futures simulated through a regional integrated assessment model under three RCP-SSP scenarios. They found that direct climate impacts, although variable across species, dominate responses for most species, especially under high-end climate change scenarios (up to 99% range loss). Land use may amplify, attenuate, or offset changes to suitable habitat extent expected from climate impact depending on species and scenario, especially in low-end climate change scenarios (RCP2.6), indicating substantial potential for targeted conservation management.

However, many bee species are ecologically and evolutionarily different from bumblebees. Kazenel et al. (2024) documented 16 years of abundance patterns for a hyper-diverse bee assemblage in a warming and drying region, linking bee declines with experimentally determined heat and desiccation tolerances, and using climate sensitivity models to project bee communities into the future. Aridity strongly predicted bee abundance for 71% of 665 bee populations. Models forecasted declines for 46% of species and predicted more homogeneous communities dominated by drought-tolerant taxa, even while total bee abundance may remain unchanged. Such community reordering could reduce pollination services, because diverse bee assemblages typically maximize pollination for plant communities.

Rahimi et al. (2024; 2025) assessed the impact of climate change on pollinator habitat suitability and pollinator-dependent crops on a global scale due to climate-dependent variables. Their findings suggested that approximately 65% of bees are likely to witness a decrease in their distribution, with climate change's impact on bees projected to be more severe in Europe (average reduction of 56%) under SSP5-8.5. Climate change's anticipated effects on bee distributions could potentially disrupt existing pollinator-plant networks, posing ecological challenges that emphasize the importance of pollinator diversity, synchrony between plants and bees, and the necessity for focused conservation efforts. Under the most pessimistic climate change scenario (SSP5-8.5), they estimate that Europe is expected to see a 4.7% decrease in area suitability for pollination-dependent crops by 2070.

Three further pan-European or gridded assessments complement the projections used here. Marshall et al. (2023) modelled the wild bee species most important for apple production in Europe under RCP2.6, RCP4.5 and RCP8.5 for 2041–2060 and 2061–2080, and found increasing spatial mismatch between key pollinating species and areas climatically suitable for apple cultivation, with widespread range contractions emerging only under the highest forcing. Millard et al. (2023) provide the only projection framed in terms of pollinator abundance rather than habitat suitability, combining the PREDICTS database with land-use and standardised temperature-anomaly interactions under RCP2.6 and RCP6.0, and estimate that pollinator abundance in cropland experiencing novel temperatures is 61% lower than in unwarmed natural habitat. At the level of the service rather than of species, the IPBES model intercomparison (Pereira et al., 2024) projects declines in regulating services including pollination under all three SSP–RCP combinations examined, and finds, in contrast to the biodiversity metrics, that the intermediate-forcing high-land-use scenario is more damaging for pollination than the highest-forcing scenario, indicating that land-use change can dominate the climate signal for this service. Finally, Newbold et al. (2025) show empirically that the response of bumblebee occurrence to recent warming interacts with local land use, being weakly negative in natural habitats but positive in human-modified ones, and that cumulative pesticide toxicity strongly reduces occurrence in natural habitats; both mechanisms are absent from the correlative SDMs used here and are relevant to interpreting our yield shocks as upper bounds.

Pressures on Pollinators and Conservation Responses

The decline of both wild and managed pollinators is intrinsically linked to the broader degradation of agro-ecosystems, where pollination represents a key ecosystem service that is subject to the same underlying multiple pressures and drivers of impact (IPBES, 2016; Potts et al., 2015).

Key pressures include:

1. Land use change – conversion of natural and semi-natural habitats to intensive agriculture
2. Habitat loss and fragmentation – resulting from urbanization and infrastructure development, removal of hedgerows and field margins, loss of semi-natural grasslands, and decline of wild flowering plants
3. Environmental pollutants – especially systemic insecticides (neonicotinoids) and fungicides
4. Invasive alien species – though impacts remain difficult to generalize

5. Climate change – affecting phenology, distributions, and species interactions

These factors are most prevalent in intensive agricultural management and monoculture cropping systems. Habitat for pollinators has been drastically reduced due to agricultural intensification, leading to loss of natural and semi-natural habitats rich in wildflowers which pollinators use for foraging, nesting and hibernating (Biesmeijer et al., 2006; Potts et al., 2010; Goulson et al., 2015; IPBES, 2016; Zattara & Aizen, 2021). While it is difficult to quantify the loss of wildflowers from agricultural landscapes, herbicides have an indirect but significant impact on wild pollinators by removing the wildflowers on which they feed. The use of pesticides – linked to agricultural intensification – also has a direct and significant impact on pollinators (Nicholson et al., 2024).

Member State surveillance of the habitats in Annex I of the EU Habitats Directive shows significant losses of pollinator habitats in the period 2013 to 2018 (EEA, 2020; Kudrnovsky et al., 2020). Over half of the grassland habitats were assessed as in unfavourable-bad conservation status and nearly half had a declining trend over the period to 2018.

Conservation potential:

Remnants of natural and semi-natural habitats, hedgerows, and field margins, which supply essential flowering and nesting resources, can become important pollinator sources in different agro-ecosystems (IPBES, 2016; Winfree et al., 2009; Morandin & Kremen, 2013; Aviron et al., 2023). However, increasing distance from field edges into crop fields greatly reduces flower visitation and the number of visiting species. On average, bee visitation rates and richness are reduced by half at distances of approximately 670 and 1,500 m, respectively, from natural vegetation (Ricketts et al., 2008). As a consequence, not only does average crop yield often decrease with distance to field margins or natural vegetation (albeit at lower rates than pollinator abundance and richness), but it also becomes less predictable (Garibaldi et al., 2011).

This spatial relationship suggests that improving measures of conservation and restoration of semi-natural habitats within agro-ecosystems can reduce pollination loss. Expert estimates indicate that restoration of pollinator habitat on at least 10% of farmland will be the minimum needed to maintain the most common wild pollinator species (Dicks et al., 2015; Martin et al., 2019; Dicks et al., 2021; EU, 2023b).

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